



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

BCSE309L – Cryptography and Network Security **Module ~ I – Fundamentals of Number Theory**

Presentation by

Dr.K.Ragavan

Assistant Professor Senior Grade I

Department of IOT

School of Computer Science and Engineering

VIT University, Vellore

COURSE OBJECTIVES

- To explore the concepts of basic number theory and cryptographic techniques.
- To impart concept of Hash and Message Authentication, Digital Signatures and authentication protocols.
- To reveal the basics of transport layer security, Web Security and various types of System Security

SYLLABUS

Module:1

Fundamentals of Number Theory

5 hours

Finite Fields and Number Theory: Modular arithmetic, Euclidian Algorithm, Primality Testing: Fermats and Eulers theorem, Chinese Remainder theorem, Discrete Logarithms

Module:2

Symmetric Encryption Algorithms

7 hours

Symmetric key cryptographic techniques: Introduction to Stream cipher, Block cipher: DES, AES, IDEA, Block Cipher Operation, Random Bit Generation and RC4.

Module:3

Asymmetric Encryption Algorithm and Key Exchange

8 hours

Asymmetric key cryptographic techniques: principles, RSA, ElGamal, Elliptic Curve cryptography, Homomorphic Encryption and Secret Sharing, Key distribution and Key exchange protocols, Diffie-Hellman Key Exchange, Man-in-the-Meddle Attack.

SYLLABUS

Module:4

Message Digest and Hash Functions

5 hours

Requirements for Hash Functions, Security of Hash Functions, Message Digest (MD5), Secure Hash Function (SHA), Birthday Attack, HMAC.

Module:5

Digital Signature and Authentication Protocols

7 hours

Authentication Requirements, Authentication Functions, Message Authentication Codes, Digital Signature Authentication, Authentication Protocols, Digital Signature Standards, RSA Digital Signature, Elgamal based Digital Signature, Authentication Applications: Kerberos, X.509 Authentication Service, Public Key Infrastructure (PKI)

SYLLABUS

Module:6	Transport Layer Security and IP Security	4 hours
-----------------	---	----------------

Transport-Layer Security, Secure Socket Layer(SSL),TLS, IP Security: Overview: IP Security Architecture, Encapsulating Payload Security

Module:7	E-mail, Web and System Security	7 hours
-----------------	--	----------------

Electronic Mail Security, Pretty Good Privacy (PGP), S/MIME, Web Security: Web Security Considerations, Secure Electronic Transaction Protocol Intruders, Intrusion Detection, Password Management, Firewalls: Firewall Design Principles, Trusted Systems.

Module:8	Contemporary Issues	2 hours
-----------------	----------------------------	----------------

COURSE OUTCOMES

On completion of this course, student should be able to:

- **To know** the fundamental mathematical concepts related to security.
- **To understand** concept of various cryptographic techniques.
- **To apprehend** the authentication and integrity process of data for various applications
- **To know** fundamentals of Transport layer security, web security, E-Mail Security and IP Security

TEXT BOOK

- Cryptography and Network Security-Principles and Practice, 8th Edition, by Stallings William, published by Pearson, 2020.

REFERENCE BOOK

- Cryptography and Network Security, 3rd Edition, by Behrouz A Forouzan and Depdeep Mukhopadhyay, published by McGrawHill, 2015.

Cryptography

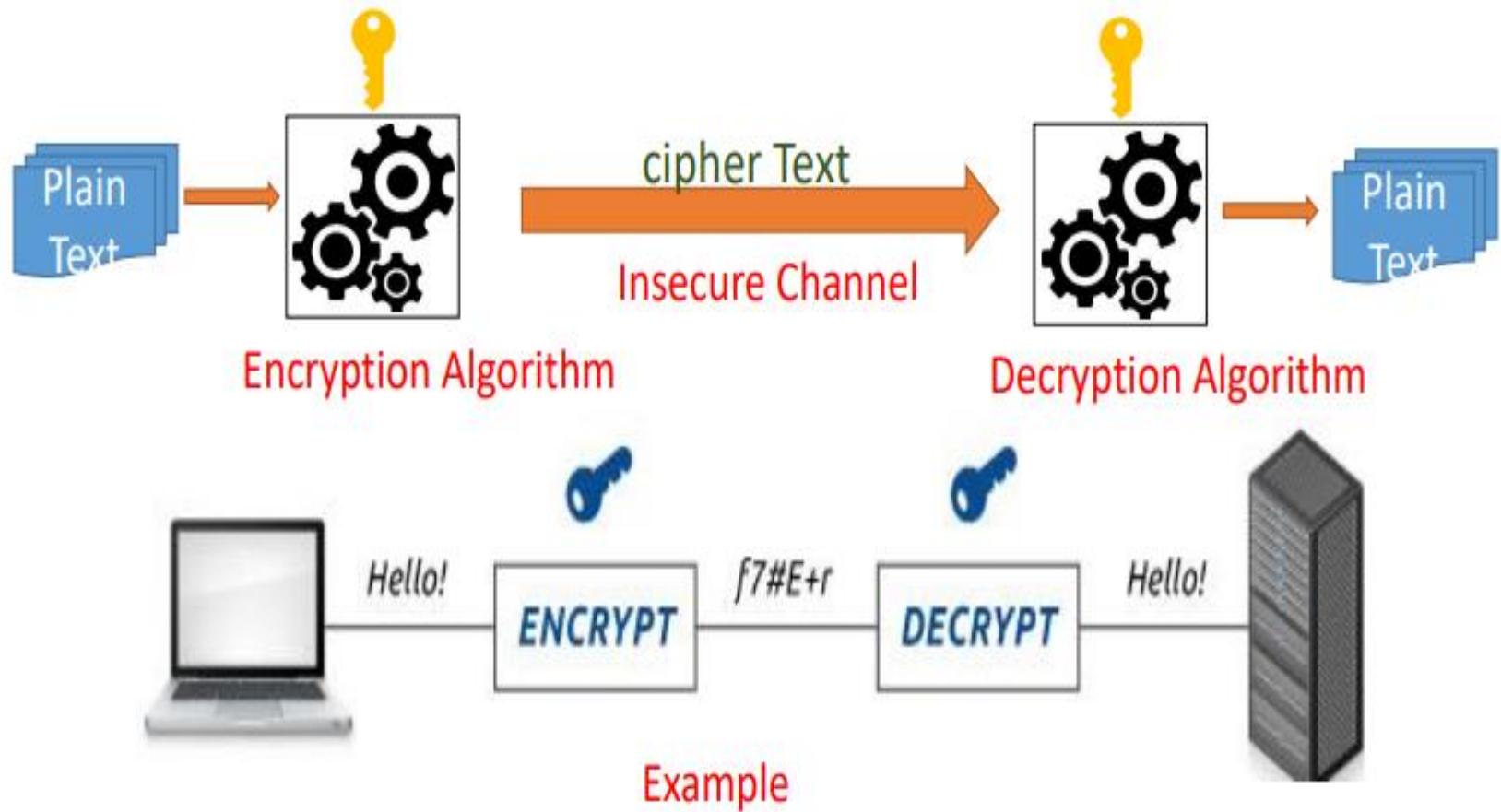
Cryptography

- It is technique of securing information and communications through use of codes. Thus preventing unauthorized access to information.
- The prefix “**crypt**” means “**hidden**” and suffix “**graphy**” means “**writing**”.
- In Cryptography, the techniques are obtained from mathematical concepts and a set of rule based calculations known as algorithms to convert messages in ways that make it hard to decode it.

Features of Cryptography (CINA)

- **Confidentiality:** Only the intended user can access the sensitive data.
- **Integrity:** Information cannot be modified in storage or during the transmission.
- **Non-repudiation:** The sender of information cannot deny his intention to send information at later stage.
- **Authentication:** The identities of sender and receiver are confirmed.

Basics of Cryptography



Most encryption/ decryption is based heavily on number theory.

Number Theory

Introduction to Number Theory

- The branch of mathematics that deals with the properties and relationships of numbers, especially the positive integers.

Basics in Number Theory:

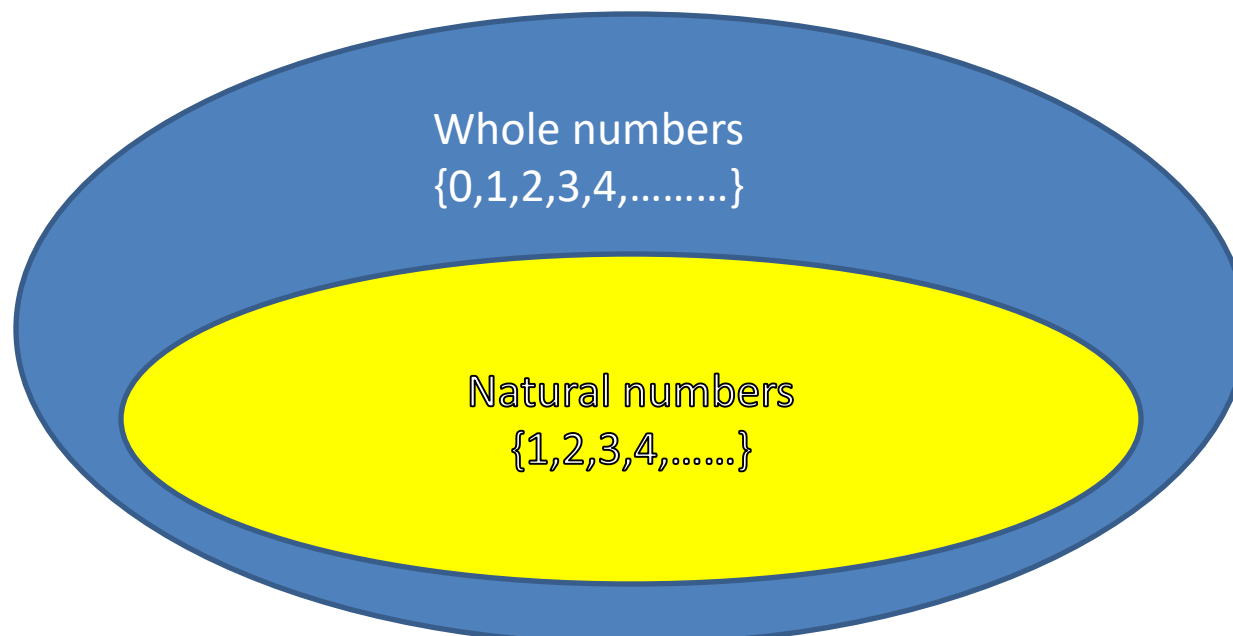
- Whole Number
- Natural Number
- Factor (Divisor)
- Prime Number
- Composite Number
- Greatest Common Divisor (Highest Common Factor)
- Co Prime (Relatively prime)

(a) Whole Number

- A number without a decimal part and it has no negatives.
- All Non Negative Integers
- $\{0,1,2,3,4,5,6,7,\dots\}$

(b) Natural Number

- Natural Number is an integer greater than 0.
- It is a member of the set $\{1,2,3,4,\dots\}$
- Natural Numbers are also known as counting numbers.



Some of the fundamental number system are:

- The natural numbers, N , (also known as non negative numbers).
- Ex: 1,2,3,4,...
- The whole numbers, W , are the set of natural numbers along with the number 0. Ex: 0,1,2,3,4...
- The integers, Z , obtained by adding negative numbers to W .
Ex: ...-2,-1,0,1,2,3,4,...
- The rational numbers, Q , obtained by adjoining reciprocals of nonzero integers to Z . Form of p/q where 'q' is not equal to zero. Ex: $1/2$, $1/5$, $3/4$
- Irrational Numbers in Square root
- The real numbers, R , any number that we can think of, except complex numbers, is a real number.

(c) Factor (Divisor)

- A factor is a number that divides another number evenly with no reminder
- Factors are numbers you multiply to get other numbers.

Example:

Factors of 24: 1,2,3,4,6,8,12,24 →

Factors of 48: 1,2,3,4,6,8,12,16, 24, 48

Factors of 32: 1, 2, 4, 8, 16, 32

Factors of 78: 1, 2, 3, 6, 13, 26, 39, and 78

$$1 \times 24 = 24$$

$$2 \times 12 = 24$$

$$3 \times 8 = 24$$

$$4 \times 6 = 24$$

~~$$5 \times \quad = 24$$~~

~~$$6 \times 4 = 24$$~~

(d) Prime Numbers

- Number greater than 1 with only two factors(divisors) - itself and one.
- It cannot be divided further by any other numbers without leaving a remainder.
- If 'N' is a prime, then the divisors are 1 and N.

(d) Prime Number - Example (Contd.....)

- 2 is a prime number.
- 3 is a prime number.
- 5 is a prime number.
- 7 is a prime number.
- 9 is not a prime number.
- 9 is a composite number.
- 33 is a composite number.

Facts about primes

(Contd.....)

- Only even prime: 2
- Smallest prime number: 2
- Is 1 a prime number? No.
- note: 1 is prime, but is generally not of interest
- Except for 2 and 5, all prime numbers end in the digit 1,3,7 or 9.
- prime numbers are central to number theory
- list of prime number less than 200 is:

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83
89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167
173 179 181 191 193 197 199

Why prime in cryptography?

(Contd.....)

- Many encryption algorithms are based on prime numbers.
- Very fast to multiply two large prime numbers.
- Extremely computer-intensive to do the reverse.
- Factoring very large prime numbers is very hard. i.e., take computers a long time.

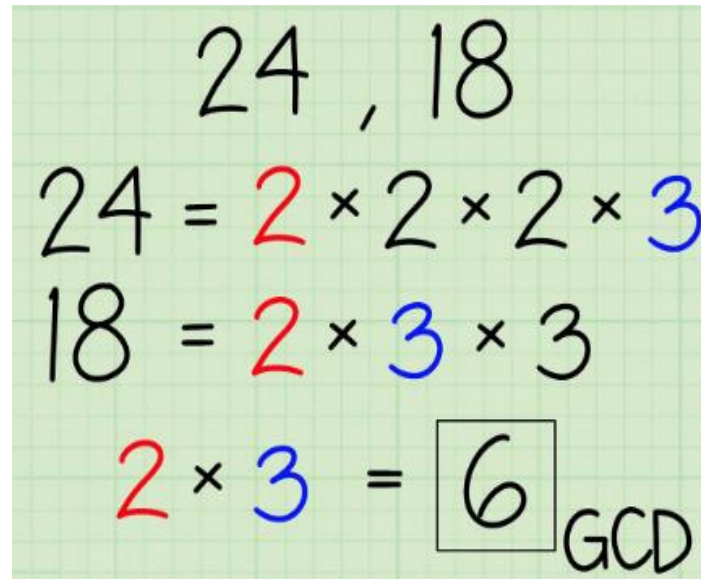
(e) Composite Number

- Numbers that have more than 2 factors are called as Composite numbers.
- Numbers that are divisible by more than two numbers.
- {4, 6, 8, 9, 10, 12, 14, 15, 16, ...}

Prime have only 2 factors: (1 and itself) 2,3,5,7,11	Composite have more than 2 factors 4,6,8,9,12,14
---	--

(f) Greatest Common Divisor

- It is also called the highest common factor (HCF)
- GCD of two numbers is the greatest number that divides both the numbers.
- Example:



Handwritten calculation on a green grid background showing the prime factorization of 24 and 18 to find their Greatest Common Divisor (GCD).

$$24, 18$$
$$24 = 2 \times 2 \times 2 \times 3$$
$$18 = 2 \times 3 \times 3$$
$$2 \times 3 = \boxed{6}_{\text{GCD}}$$

Integer Division

- In integer arithmetic, if we divide **a** by **n** (**a/n**), we can get **q** and **r**.
- The relationship between these four integers can be shown as,

$$a = q \times n + r$$

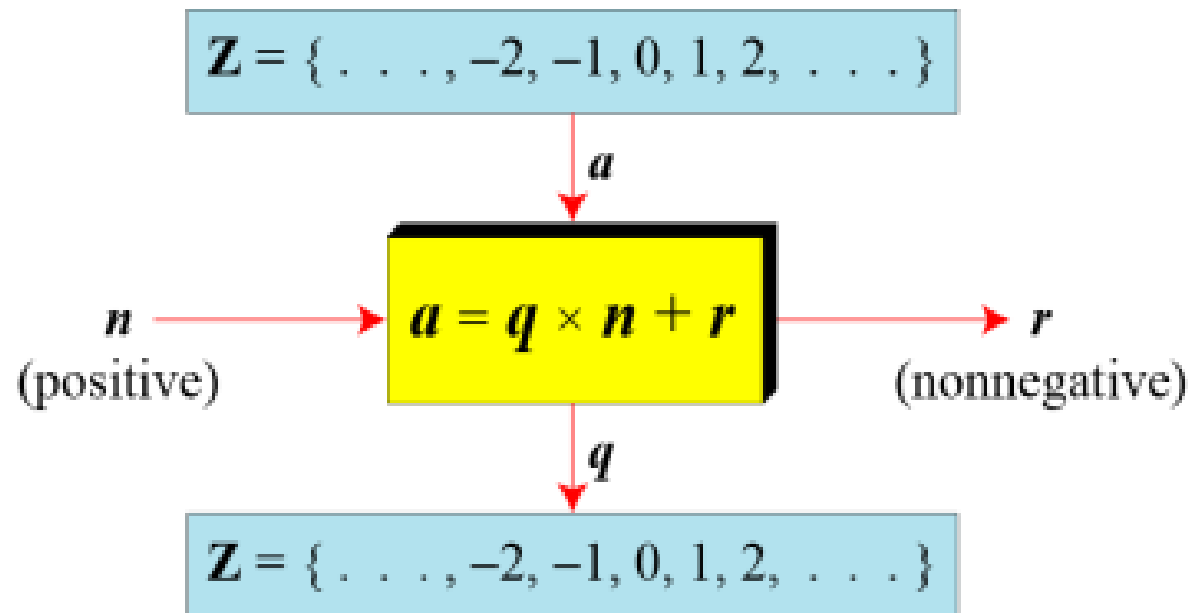
- Assume that **a = 255** and **n = 11**.
- We can find **q = 23** and **r = 2** using the division algorithm.

$$\begin{array}{r} 23 \leftarrow q \\ \hline 11 \overline{) 255} \\ \underline{22} \\ 35 \\ \underline{33} \\ 2 \leftarrow r \end{array}$$

(Note: In the original image, 'n' points to '11' and 'a' points to '255')

Integer Division / Division Algorithm

- Division algorithm for integers



Integer Division

- Assume that $a = -255$ and $n = 11$.
- We can find $q = -23$ and $r = -2$ using the division algorithm.

$$\begin{array}{r}
 \begin{array}{c} n \longrightarrow 11 \end{array} \left| \begin{array}{r} -23 \longleftarrow q \\ -255 \longleftarrow a \\ + \\ -22 \\ \hline -35 \\ 33 \\ \hline -2 \longleftarrow r \end{array} \right.
 \end{array}$$

To convert a negative remainder into non negative remainder do the following,

- Decrement the value of q by 1.
- Add the value of n to r to make r positive.

$$q = q - 1$$

$$r = r + n$$

Integer Division

- Assume that $a = -255$ and $n = 11$.
- We can find $q = -23$ and $r = -2$ using the division algorithm.

$$\begin{array}{r}
 \begin{array}{c} n \longrightarrow 11 \end{array} \quad \begin{array}{r}
 \begin{array}{r}
 -23 \longleftarrow q \\
 \hline
 -255 \longleftarrow a \\
 + \\
 -22 \\
 \hline
 -35 \\
 33 \\
 \hline
 -2 \longleftarrow r
 \end{array}
 \end{array}
 \end{array}$$

Conversion:

$$q = q - 1$$

$$r = r + n$$

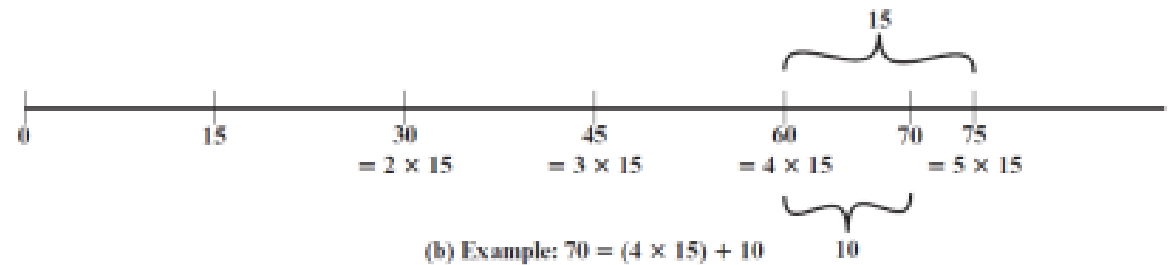
$$q = -23 \rightarrow -23 - 1 = -24$$

$$r = -2 \rightarrow -2 + 11 = 9$$

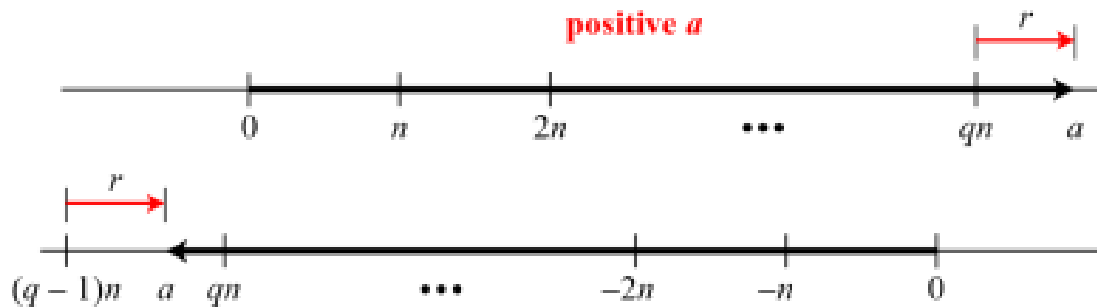
$$a = (q \times n) + r$$

$$-255 = (-24 \times 11) + 9$$

Integer Division



Case of
positive a



Case of
negative a

Divisibility

- If a is not zero and we let $r = 0$ in the division relation, we get

$$a = q \times n$$

- If the remainder is zero, $n|a$
- If the remainder is not zero, $n \nmid a$

NOTE

- $n|a$
 - n divides a
 - n is a divisor of a

Divisibility Example

- $32/4$
 - $a=32$
 - $n=4$
 - **Relationship:** $32 = (8 \times 4) + 0$
 - $q=8$; $r=0$
 - **Representation:** $4|32$
 - 4 divides 32
 - 4 is a divisor of 32
- $42/8$
 - $a=42$
 - $n=8$
 - **Relationship:** $42 = (5 \times 8) + 2$
 - $q=5$; $r=2$
 - **Representation:** $8 \nmid 42$
 - 8 does not divide 42
 - 8 is not a divisor of 42

Properties of Divisibility

- If $a|1$, then $a = \pm 1$.
 - If $a|b$ and $b|a$, then $a = \pm b$.
 - Any $b \neq 0$ divides 0.
 - If $a|b$ and $b|c$, then $a|c$:
1. Divisor of 1 is either +1 or -1.
 2. A number divides itself (+ or -)
 3. 0 has no divisor but 0 is a dividend of all numbers.
 - $25/0 \rightarrow$ not defined
 - $0/25 \rightarrow 0$
 - $0|25$
 4. $11|66$ and $66|198$
 - $11|198$

Greatest Common Divisor (GCD)

- The greatest common divisor of two positive integers is the **largest integer that can divide both integers**.
- $\text{GCD}(a,b)$ of a and b is the largest integer that divides evenly into both a and b .

$$\text{eg } \text{GCD}(60,24) = 12$$

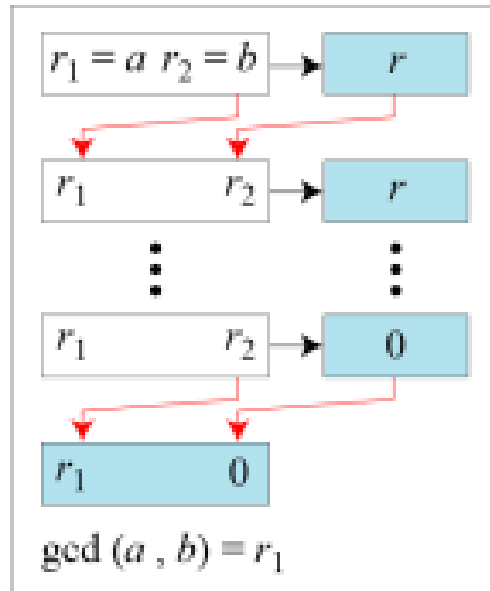
- We define $\text{gcd}(0, 0) = 0$
- We define $\text{gcd}(a,0) = |a|$
- When $\text{gcd}(a,b)=1$, it implies that there are no common divisor except 1.
 - So, a and b are said to be **relatively prime** to each other.
 - Eg: $\text{gcd}(8,15)=1$

How to find GCD by division
method?

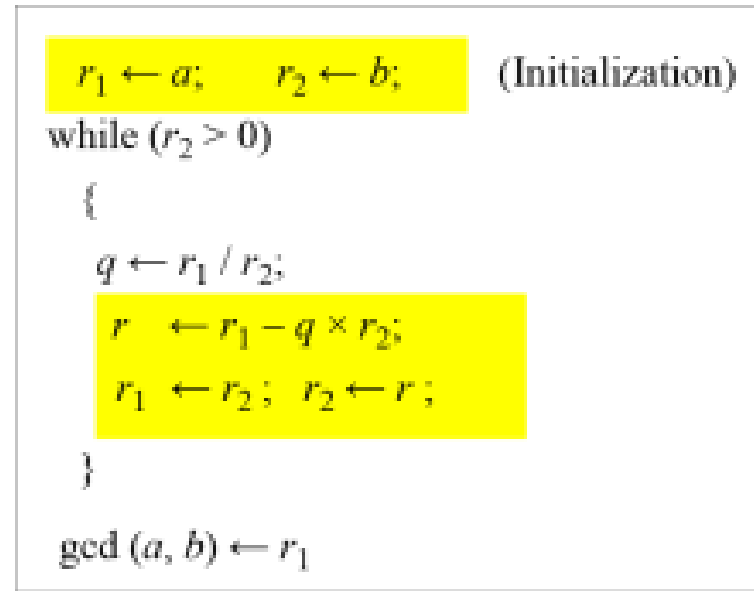
GCD by Division Method (Euclidean Algorithm)

- **Step 1:** Divide the large number by the smaller one.
- **Step 2:** Then the remainder is treated as divisor and the divisor as dividend.
- **Step 3:** Divide the first divisor by the first remainder.
- **Step 4:** Divide the second divisor by the second remainder.
- **Step 5:** Continue this process till the remainder becomes 0.
- **Step 6:** The last divisor is the required highest common factor (H.C.F) of the given numbers.

Euclidean Algorithm



a. Process



b. Algorithm

When $\text{gcd}(a, b) = 1$, we say that a and b are relatively prime.

Example - 1

Step 1:
$$\begin{array}{r} 1 \\ 18 \overline{) 30} \\ \underline{-18} \quad 1 \end{array}$$
 1st divisor 1st remainder

Step 2:
$$\begin{array}{r} 2 \\ 12 \overline{) 18} \\ \underline{-12} \quad 6 \end{array}$$
 Divide the 1st divisor by the 1st remainder.

Step 3:
$$\begin{array}{r} 3 \\ 6 \overline{) 12} \\ \underline{-12} \quad 0 \end{array}$$
 Divide the 2nd divisor by the 2nd remainder.

Highest Common Factor
$$\begin{array}{r} 6 \\ 6 \overline{) 12} \\ \underline{-12} \quad 0 \end{array}$$
 Remainder is 0.

$$\text{GCD}(30, 18) = 6$$

Example - 2

Step 1:
$$\begin{array}{r} 2 \\ 75 \overline{) 180} \\ \underline{-150} 2 \end{array}$$

Step 2:
$$\begin{array}{r} 2 \\ 75 \overline{) 180} \\ \underline{-150} 2 \\ 30 \overline{) 75} \end{array}$$

Step 3:
$$\begin{array}{r} 2 \\ 75 \overline{) 180} \\ \underline{-150} 2 \\ 30 \overline{) 75} \\ \underline{-60} 2 \\ 15 \overline{) 30} \\ \underline{-30} \\ 0 \end{array}$$

$$\text{GCD}(75, 180) = 15$$

GCD by Division Method - Exercise

- Find the GCD of 30 and 45
- Find the GCD of 442 and 546
- $\text{GCD}(1970, 1066) = ?$

(g) Co-Prime (relatively prime) Number

- Two numbers are said to be relatively prime, if they have no prime factors in common, and their only common factor is 1.
- If $\text{GCD}(a,b)=1$, then 'a' and 'b' are relatively prime numbers.
- NOTE: It is not necessary that a,b to be a prime numbers.

Relatively Prime Numbers

Question 1: Are 4 and 13 relatively prime?

Solution:

	4	13
Divisors	1, 2, 4	1, 13
Common Divisors	1	
Greatest Common Divisor (GCD)	1	

$$\text{GCD}(4, 13) = 1$$

Yes, 4 and 13 are relatively prime numbers.

Co-prime - Example

(Contd.....)

Question 2: Are 15 and 21 relatively prime?

Solution:

	15	21
Divisors	1, 3, 5, 15	1, 3, 7, 21
Common Divisors	1, 3	
Greatest Common Divisor (GCD)	3	

$$\text{GCD}(15, 21) = 3$$

No, 15 and 21 are not relatively prime numbers.

Co-prime

(Contd.....)

- Need for finding GCD: To ensure whether the numbers are relatively prime or not.
- NOTE: apply Euclid's algorithm to find GCD of numbers.

Relatively Prime Numbers

a	b	GCD(a, b)	Relatively Prime?	Remarks
11	17	1	Yes	'a' and 'b' are prime
11	21	1	Yes	'a' is prime and 'b' is composite
12	77	1	Yes	'a' and 'b' are composite

Find the GCD(790, 121) using Euclid's algorithm and determine whether they are relatively prime or not.

Properties of Co-Prime Numbers

- Set of numbers that have only one common factor, namely 1.
or when they have their greatest common factor (GCD) as 1.
- Prime numbers are co-prime to each other.
- Any two successive numbers are always co-prime.
- The sum of any two co-prime numbers is always co-prime with their product.

Homework

Identify the co-prime numbers

- 24, 36
- 20, 21
- 542, 446

Euclidean Algorithm

- Find the greatest common divisor of 25 and 60.

q	r1	r2	r
0	25	60	25

Divide $25/60 = 0.4166$

$q = 0$

$r = 25 \ (0.416 \cdot 60)$

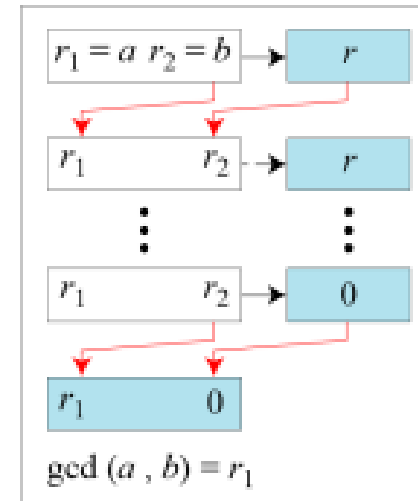
Euclidean Algorithm

- Find the greatest common divisor of 25 and 60.

q	r1	r2	r
0	25	60	25
2	60	25	10

Divide $60/25 = 2.4$

$q = 2$
 $r = 10 \text{ (} 0.4 \cdot 25 \text{)}$



Process

q	r1	r2	r
0	25	60	25
2	60	25	10
2	25	10	5

Divide $25/10 = 2.5$

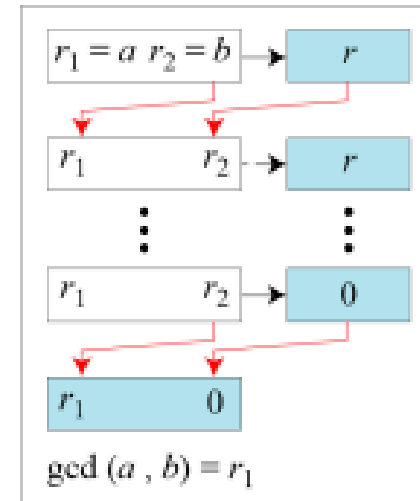
$q = 2$
 $r = 5 \text{ (} 0.5 \cdot 10 \text{)}$

Find the greatest common divisor of 25 and 60.

q	r1	r2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0

Divide $10/5 = 2$

q = 2
r = 0



Process

q	r1	r2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

When r2=0

Stop

Euclidean Algorithm

- Find the greatest common divisor of 25 and 60.

q	r1	r2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

GCD(25,60)=5

Euclidean Algorithm

- Find the greatest common divisor of 2740 and 1760.

q	r1	r2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

$$\text{GCD}(2740, 1760) = 20$$

Euclidean Algorithm

• $\gcd(1160718174, 316258250)=?$

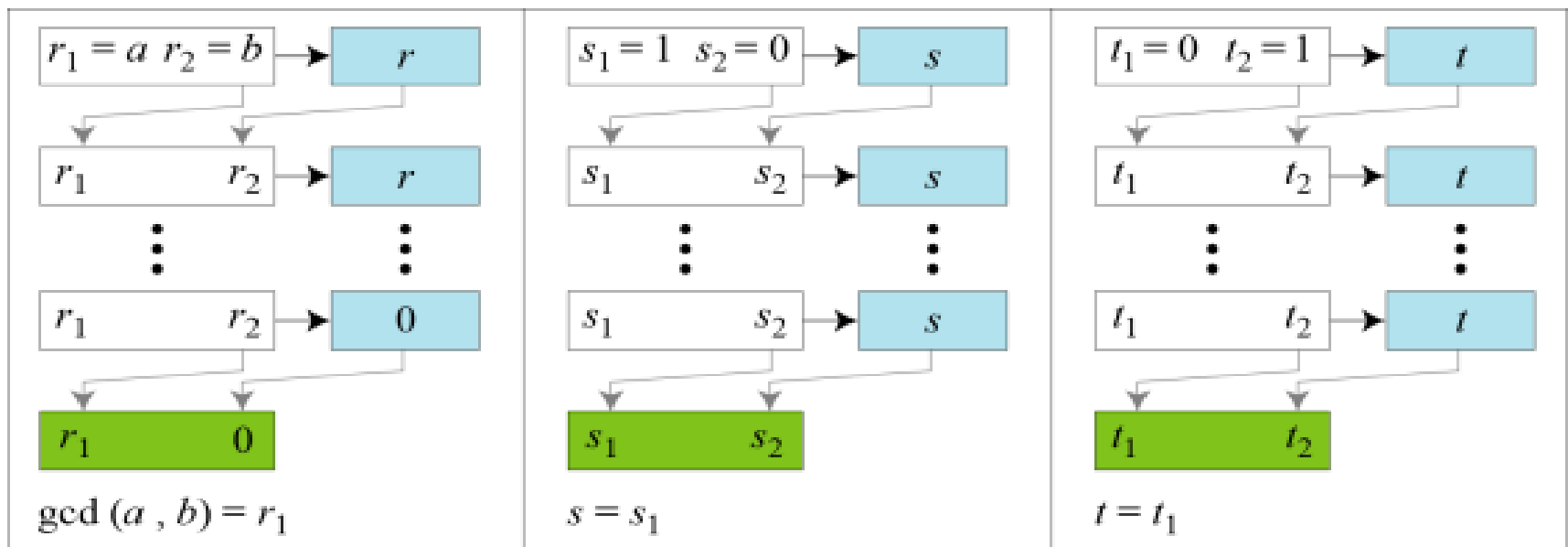
q	r1	r2	r
3	1160718174	316258250	211943424
1	316258250	211943424	104314826
2	211943424	104314826	3313772
31	104314826	3313772	1587894
2	3313772	1587894	137984
11	1587894	137984	70070
1	137984	70070	67914
1	70070	67914	2156
31	67914	2156	1078
2	2156	1078	0
	1078	0	

Extended Euclidean Algorithm

- For given integers a and b , the extended Euclidean algorithm not only calculates the **greatest common divisor d** but also **two additional integers s and t** that satisfy the following equation.

$$a.s + b.t = d = \gcd(a, b)$$

- s and t will have opposite signs.
- Eg: $\gcd(161, 28) = 7$, $s = -1$ and $t = 6$



a. Process

Extended Euclidean Algorithm

```
 $r_1 \leftarrow a; \quad r_2 \leftarrow b;$   
 $s_1 \leftarrow 1; \quad s_2 \leftarrow 0;$   
 $t_1 \leftarrow 0; \quad t_2 \leftarrow 1;$ 
```

(Initialization)

```
while ( $r_2 > 0$ )
```

```
{
```

```
   $q \leftarrow r_1 / r_2;$ 
```

```
   $r \leftarrow r_1 - q \times r_2;$ 
```

```
   $r_1 \leftarrow r_2; \quad r_2 \leftarrow r;$ 
```

(Updating r 's)

```
   $s \leftarrow s_1 - q \times s_2;$ 
```

```
   $s_1 \leftarrow s_2; \quad s_2 \leftarrow s;$ 
```

(Updating s 's)

```
   $t \leftarrow t_1 - q \times t_2;$ 
```

```
   $t_1 \leftarrow t_2; \quad t_2 \leftarrow t;$ 
```

(Updating t 's)

```
}
```

```
gcd( $a, b$ )  $\leftarrow r_1; \quad s \leftarrow s_1; \quad t \leftarrow t_1$ 
```

b. Algorithm

Extended Euclidean Algorithm

$$s \leftarrow s_1 - q \times s_2;$$

(Updating s 's)

$$s_1 \leftarrow s_2; s_2 \leftarrow s;$$

$$t \leftarrow t_1 - q \times t_2;$$

(Updating t 's)

$$t_1 \leftarrow t_2; t_2 \leftarrow t;$$

- Given $a = 161$ and $b = 28$, find $\gcd(a, b)$ and the values of s and t .

q	r1	r2	r	s1	s2	s	t1	t2	t
5	161	28	21	1	0	1	0	1	-5

$a=161$ $b=28$

$$161/28 = 5.75$$

$$q = 5$$

$$r = 21$$

$$s = s_1 - (q \times s_2)$$

$$s = 1 - (5 \times 0)$$

$$s = 1$$

$$t = t_1 - (q \times t_2)$$

$$t = 0 - (5 \times 1)$$

$$t = -5$$

Extended Euclidean Algorithm

$$s \leftarrow s_1 - q \times s_2;$$

$$s_1 \leftarrow s_2; s_2 \leftarrow s;$$

(Updating s 's)

$$t \leftarrow t_1 - q \times t_2;$$

$$t_1 \leftarrow t_2; t_2 \leftarrow t;$$

(Updating t 's)

- Given $a = 161$ and $b = 28$, find $\gcd(a, b)$ and the values of s and t .

q	r1	r2	r	s1	s2	s	t1	t2	t
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6

$a=28$ $b=21$

28/21 = 1.333

$q = 1$

$r = 7$

$s = s_1 - (q \times s_2)$

$s = 0 - (1 \times 1)$

$s = -1$

$t = t_1 - (q \times t_2)$

$t = 1 - (1 \times -5)$

$t = 6$

Extended Euclidean Algorithm

$$s \leftarrow s_1 - q \times s_2;$$

(Updating s 's)

$$s_1 \leftarrow s_2; s_2 \leftarrow s;$$

$$t \leftarrow t_1 - q \times t_2;$$

(Updating t 's)

$$t_1 \leftarrow t_2; t_2 \leftarrow t;$$

- Given $a = 161$ and $b = 28$, find $\gcd(a, b)$ and the values of s and t .

q	r1	r2	r	s1	s2	s	t1	t2	t
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	7	0		-1	4		6	-23	

Stop when r_2 becomes 0

GCD(161,28)=7

And the two integers are as follows,

$$s = -1$$

$$t = 6$$

$$s \cdot a + t \cdot b = \text{GCD}(a, b)$$

$$(-1 \cdot 161) + (6 \cdot 28) = 7$$

$$-161 + 168 = 7$$

$$7 = 7$$

Extended Euclidean Algorithm

$$s \leftarrow s_1 - q \times s_2;$$

$$s_1 \leftarrow s_2; s_2 \leftarrow s;$$

(Updating s 's)

$$t \leftarrow t_1 - q \times t_2;$$

$$t_1 \leftarrow t_2; t_2 \leftarrow t;$$

(Updating t 's)

- Given $a = 17$ and $b = 0$, find $\gcd(a, b)$ and the values of s and t .

q	r1	r2	r	s1	s2	s	t1	t2	t
	17	0		1	0		0	1	

Stop when r_2 becomes 0

 $\gcd(17, 0) = 17$

And the two integers are as follows,

$$s = 1$$

$$t = 0$$

 $s \cdot a + t \cdot b = \gcd(a, b)$

$$(1 \cdot 17) + (0 \cdot 0) = 17$$

$$17 + 0 = 17$$

$$17 = 17$$

Extended Euclidean Algorithm

$$s \leftarrow s_1 - q \times s_2;$$

(Updating s 's)

$$s_1 \leftarrow s_2; s_2 \leftarrow s;$$

$$t \leftarrow t_1 - q \times t_2;$$

(Updating t 's)

$$t_1 \leftarrow t_2; t_2 \leftarrow t;$$

- Given $a = 0$ and $b = 45$, find $\gcd(a, b)$ and the values of s and t .

q	r1	r2	r	s1	s2	s	t1	t2	t
0	0	45	0	1	0	1	0	1	0

$a=0$ $b=45$

$$0/45 = 0$$

$$q=0$$

$$r=0$$

$$s=s_1-(q \times s_2)$$

$$s=1-(0 \times 0)$$

$$s=1$$

$$t=t_1-(q \times t_2)$$

$$t=0-(0 \times 1)$$

$$t=0$$

Extended Euclidean Algorithm

$$s \leftarrow s_1 - q \times s_2;$$

$$s_1 \leftarrow s_2; s_2 \leftarrow s;$$

(Updating s 's)

$$t \leftarrow t_1 - q \times t_2;$$

$$t_1 \leftarrow t_2; t_2 \leftarrow t;$$

(Updating t 's)

- Given $a = 0$ and $b = 45$, find $\gcd(a, b)$ and the values of s and t .

q	r1	r2	r	s1	s2	s	t1	t2	t
0	0	45	0	1	0	1	0	1	0
	45	0		0	1		1	0	

Stop when r_2 becomes 0

GCD(0,45)=45

And the two integers are as follows,

$$s=0$$

$$t=1$$

$$s*a + t*b = \text{GCD}(a,b)$$

$$(0*0) + (1*45) = 45$$

$$0 + 45 = 45$$

$$45 = 45$$

Homework

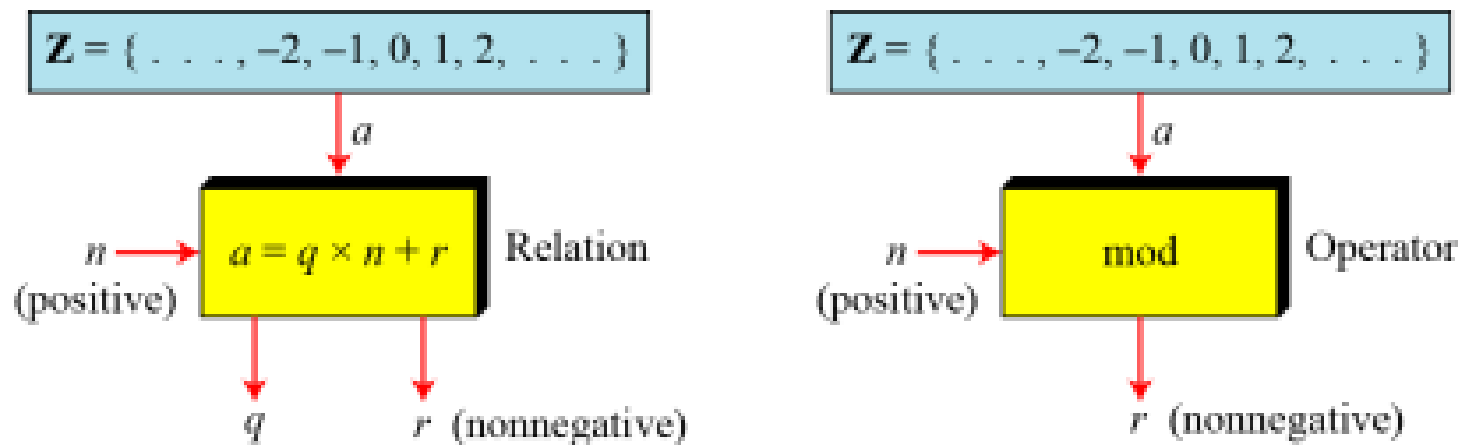
- Given $a = 1759$ and $b = 550$, find $\gcd(a, b)$ and the values of x and y .
 - Answer: $\text{GCD}(1759, 550) = 1$; $x = -111$; $y = 355$

Modular Arithmetic

- The division relationship $(a = q \times n + r)$ discussed in the previous section has **two inputs (a and n)** and **two outputs (q and r)**.
- In **modular arithmetic**, we are **interested in** only one of the outputs, the **remainder r**.
- **Topics discussed in this section:**
 - Modular Operator
 - Set of Residues
 - Congruence
 - Operations in Z_n
 - Addition and Multiplication Tables
 - Different Sets

Modulo Operator

- The modulo operator is shown as **mod**.
- The second input (**n**) is called the **modulus**.
- The output **r** is called the **residue**.



Division algorithm and modulo operator

Modulo Operator

- Find the result of the following operations:

- **$27 \bmod 5 = 2$**

- $27/5 = 5.4$
- $q=5$
- $r=(5.4-5)*5=2$

- **$36 \bmod 12 = 0$**

- $36/12 = 3 \rightarrow r=0$

- **$-18 \bmod 14 = 10$**

- $-18/14 = -1.2857$
- $q = -1$
- **$r = (-1.2857 + 1) * 14 = -4$**
- Decrement q by 1 and add n to r.
 $q = -2$; $r = -4 + 14 = 10$

- **$-7 \bmod 10 = 3$**

- $-7/10 = -0.7$
- $q = 0$
- **$r = (-0.7 * 10) = -7$**
- Decrement q by 1 and add n to r.
 $q = -1$; $r = -7 + 10 = 3$

Set of Residues

- The modulo operation creates a set, which in modular arithmetic is referred to as **the set of least residues modulo n , or Z_n** .

$$Z_n = \{ 0, 1, 2, 3, \dots, (n-1) \}$$

$$Z_2 = \{ 0, 1 \}$$

$$Z_6 = \{ 0, 1, 2, 3, 4, 5 \}$$

$$Z_{11} = \{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \}$$

Some Z_n sets

Congruence

- **Congruence**: The quality or **state of agreeing, coinciding**
- Two integers a and b are said to be congruent modulo n , if
$$(a \bmod n) = (b \bmod n)$$
- This is written as $a \equiv b \pmod{n}$

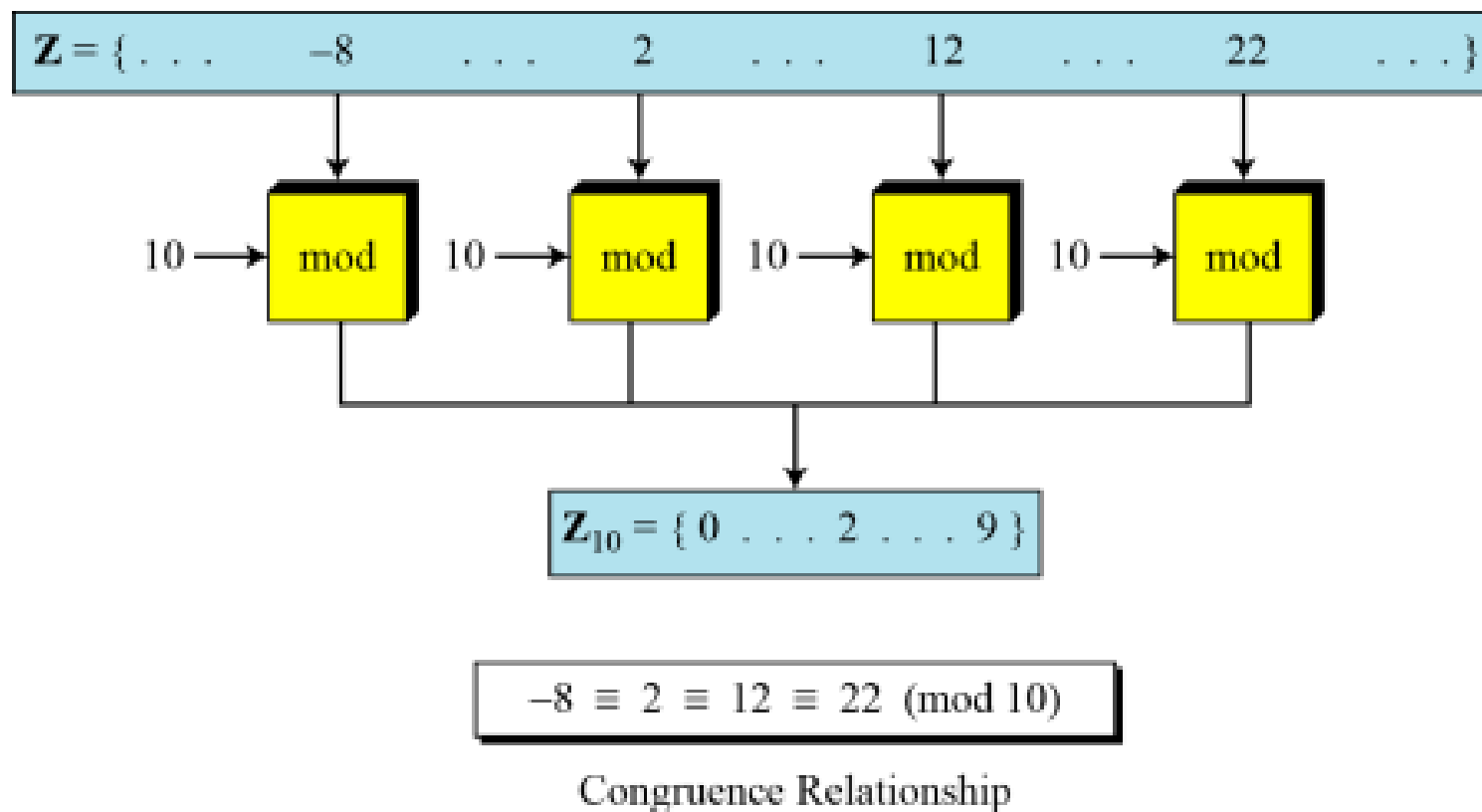
$$2 \equiv 12 \pmod{10}$$

$$3 \equiv 8 \pmod{5}$$

$$13 \equiv 23 \pmod{10}$$

$$8 \equiv 13 \pmod{5}$$

Congruence



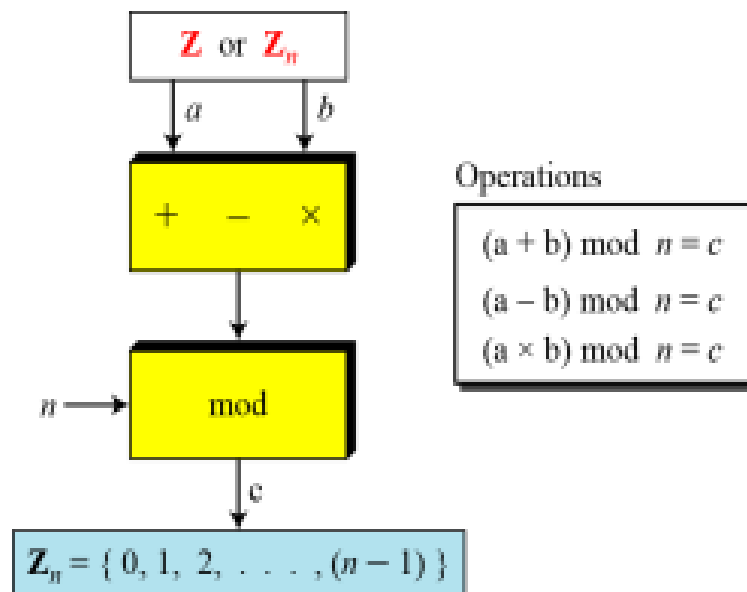
Properties of Congruences

1. $a \equiv b \pmod{n}$ if $n \mid (a - b)$.
 2. $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$.
 3. $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv c \pmod{n}$.
-

$23 \equiv 8 \pmod{5}$	because	$23 - 8 = 15 = 5 \times 3$
$-11 \equiv 5 \pmod{8}$	because	$-11 - 5 = -16 = 8 \times (-2)$
$81 \equiv 0 \pmod{27}$	because	$81 - 0 = 81 = 27 \times 3$

Operation in Z_n

- The **three binary operations** that we discussed for the set Z can also be defined for the set Z_n .
- The result may need to be mapped to Z_n using the mod operator.



Operation in Z_n

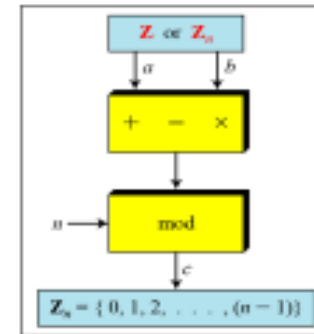
Perform the following operations (the inputs come from Z_n):

- **Add 7 to 14 in Z_{15}**
 - $(7+14) \bmod 15 = 21 \bmod 15 = 6$
- **Subtract 11 from 7 in Z_{13}**
 - $(7-11) \bmod 13 = -4 \bmod 13 = 9 \bmod 13 = 9$
- **Multiply 11 by 7 in Z_{20}**
 - $(11*7) \bmod 20 = 77 \bmod 20 = 17$

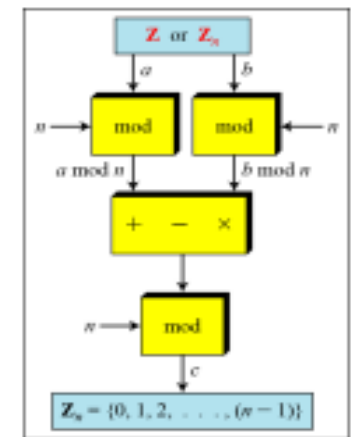
Operation in Z_n

Properties

1. $[(a \bmod n) + (b \bmod n)] \bmod n = (a + b) \bmod n$
2. $[(a \bmod n) - (b \bmod n)] \bmod n = (a - b) \bmod n$
3. $[(a \bmod n) \times (b \bmod n)] \bmod n = (a \times b) \bmod n$



a. Original process



b. Applying properties

Example

1. $[(11 \bmod 8) + (15 \bmod 8)] \bmod 8 = 10 \bmod 8 = 2$; $(11 + 15) \bmod 8 = 26 \bmod 8 = 2$
2. $[(11 \bmod 8) - (15 \bmod 8)] \bmod 8 = -4 \bmod 8 = 4$; $(11 - 15) \bmod 8 = -4 \bmod 8 = 4$
3. $[(11 \bmod 8) \times (15 \bmod 8)] \bmod 8 = 21 \bmod 8 = 5$; $(11 \times 15) \bmod 8 = 165 \bmod 8 = 5$

- In arithmetic, we often need to find the remainder of powers of 10 when divided by an integer.

$$10^n \bmod x = (10 \bmod x)^n \quad \text{Applying the third property } n \text{ times.}$$

$$10 \bmod 3 = 1 \quad \rightarrow \quad 10^n \bmod 3 = (10 \bmod 3)^n = 1$$

$$10 \bmod 9 = 1 \quad \rightarrow \quad 10^n \bmod 9 = (10 \bmod 9)^n = 1$$

$$10 \bmod 7 = 3 \quad \rightarrow \quad 10^n \bmod 7 = (10 \bmod 7)^n = 3^n \bmod 7$$

Inverses

- When we are working in modular arithmetic, we often need to find the inverse of a number relative to an operation.

We are normally looking for

- An **additive inverse** (relative to an addition operation) or
- A **multiplicative inverse** (relative to a multiplication operation).

Inverses: Additive

- In Z_n , two numbers a and b are additive inverses of each other if

$$a + b \equiv 0 \pmod{n}$$

In modular arithmetic, each integer has an additive inverse. The sum of an integer and its additive inverse is congruent to 0 modulo n.

Inverses: Additive

- Find all additive inverse pairs in Z_{10} .

Solution:

- $Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- Consider '0' the additive inverse is itself '0' $[0+0] \equiv 0 \pmod{10}$
- '1' additive inverse is '9', $[1+9] \equiv 0 \pmod{10}$
- 2 additive inverse is 8, $[2+8] \equiv 0 \pmod{10}$
- The six pairs of additive inverses are
 $(0, 0), (1, 9), (2, 8), (3, 7), (4, 6),$ and $(5, 5).$

Inverses: Multiplicative

- In Z_n , two numbers a and b are the multiplicative inverse of each other if

$$a \times b \equiv 1 \pmod{n}$$

In modular arithmetic, an integer may or may not have a multiplicative inverse.

When it does, the product of the integer and its multiplicative inverse is congruent to **1 modulo n .**

Inverses: Multiplicative

- Find all multiplicative inverses in Z_{10} .

Solution

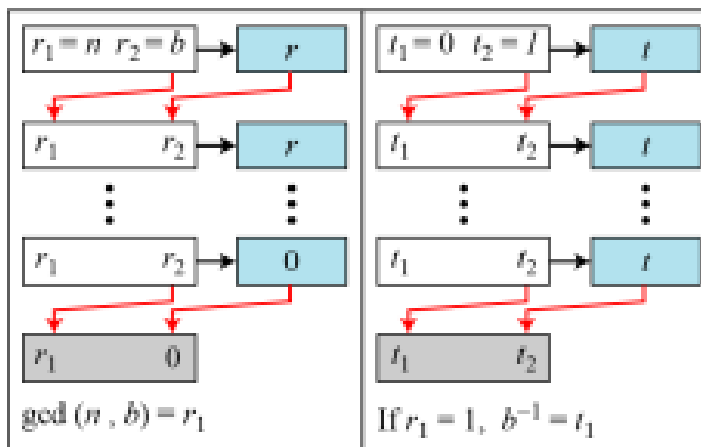
- $Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $0 * \text{NA} \equiv 1 \pmod{10}$
 - 0 has no multiplicative inverse under modulo 10.
- $1 * ? \equiv 1 \pmod{10}$
 - $1 * 1 \equiv 1 \pmod{10}$
 - The multiplicative inverse of 1 is 1 under modulo 10.
- $2 * ? \equiv 1 \pmod{10}$
 - 2 has no multiplicative inverse under modulo 10.
- $3 * ? \equiv 1 \pmod{10}$
 - $3 * 7 \equiv 1 \pmod{10}$
 - The multiplicative inverse of 3 is 7 under modulo 10.
- There are only three pairs: **(1, 1), (3, 7) and (9, 9)**.
- The numbers 0, 2, 4, 5, 6, and 8 do not have a multiplicative inverse.

Inverses: Multiplicative

- Find the multiplicative inverse of 8 in Z_{10} .

Solution

- Multiplicative inverse for a number only exists if the number and the modulo operator are relatively prime to each other.
- There is no multiplicative inverse because $\gcd(10, 8) = 2 \neq 1$.
- Using extended Euclidean algorithm to find multiplicative inverse



a. Process

b. Algorithm

Inverses: Multiplicative

- Find the multiplicative inverse of 11 in $\mathbb{Z}_{26}$.
- Solution**

q	r_1	r_2	r	t_1	t_2	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
	1	0		-7	26	

- The gcd (26, 11) is 1; the inverse of 11 is -7 or 19

```

 $r_1 \leftarrow a; \quad r_2 \leftarrow b;$ 
 $t_1 \leftarrow 0; \quad t_2 \leftarrow 1;$ 

while ( $r_2 > 0$ )
{
     $q \leftarrow r_1 / r_2;$ 

     $r \leftarrow r_1 - q \times r_2;$ 
     $r_1 \leftarrow r_2; \quad r_2 \leftarrow r;$ 

     $t \leftarrow t_1 - q \times t_2;$ 
     $t_1 \leftarrow t_2; \quad t_2 \leftarrow t;$ 
}

if ( $r_1 = 1$ ) then  $b^{-1} \leftarrow t_1$ 
    
```

b. Algorithm

- Find the multiplicative inverse of 23 in $\mathbb{Z}_{100}$.

- Solution**

q	r_1	r_2	r	t_1	t_2	t
4	100	23	8	0	1	-4
2	23	8	7	1	-4	19
1	8	7	1	-4	9	-13
7	7	1	0	9	-13	100
	1	0		-13	100	

- The gcd (100, 23) is 1; the inverse of 23 is -13 or 87.

```

 $r_1 \leftarrow a; \quad r_2 \leftarrow b;$ 
 $t_1 \leftarrow 0; \quad t_2 \leftarrow 1;$ 

```

```

while ( $r_2 > 0$ )

```

```

{
   $q \leftarrow r_1 / r_2;$ 

```

```

   $r \leftarrow r_1 - q \times r_2;$ 

```

```

   $r_1 \leftarrow r_2; \quad r_2 \leftarrow r;$ 

```

```

   $t \leftarrow t_1 - q \times t_2;$ 

```

```

   $t_1 \leftarrow t_2; \quad t_2 \leftarrow t;$ 

```

```

}

```

```

if ( $r_1 = 1$ ) then  $b^{-1} \leftarrow t_1$ 

```

b. Algorithm

- Find the inverse of 12 in $\mathbb{Z}_{26}$.

- Solution**

q	r_1	r_2	r	t_1	t_2	t
2	26	12	2	0	1	-2
6	12	2	0	1	-2	13
	2	0		-2	13	

- The gcd (26, 12) is 2; the inverse does not exist.

Addition and Multiplication Tables

	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2
4	4	5	6	7	8	9	0	1	2	3
5	5	6	7	8	9	0	1	2	3	4
6	6	7	8	9	0	1	2	3	4	5
7	7	8	9	0	1	2	3	4	5	6
8	8	9	0	1	2	3	4	5	6	7
9	9	0	1	2	3	4	5	6	7	8

Addition Table in $\mathbf{Z}_{10}$

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	0	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

Multiplication Table in $\mathbf{Z}_{10}$

Different Sets

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

$$\mathbb{Z}_6^* = \{1, 5\}$$

$$\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$$

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$$

$$\mathbb{Z}_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathbb{Z}_{10}^* = \{1, 3, 7, 9\}$$

**We need to use $\mathbb{Z}_n$ when additive inverses are needed;
we need to use $\mathbb{Z}_n^*$ when multiplicative inverses are needed.**